

# Estimating Deaton’s unit-value model: the `duvm` Stata module

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Software: `duvm` 1.0.0, <https://github.com/aabddd12/duvm>

## Abstract

`duvm` estimates the unit-value model of Deaton (1988, 1990, 1997): own- and cross-price elasticities of quantity, expenditure elasticities and quality elasticities for a group of goods, from the budget shares and unit values that households report, using the spatial variation of prices across clusters. The estimator is Deaton’s two-stage procedure, closed-form and written in Mata: within-cluster regressions, between-cluster regressions of the cluster averages with the errors-in-variables correction, the quality parameter derived from weak separability, completion of the system with a composite good, and an approximate Slutsky symmetry. `duvm` adds what the published programs lack: sampling weights with the effective cluster sizes they imply; a linearized variance built from the influence functions of every first- and second-stage moment, including the generated-regressor terms of the first stage, combined through the analytic Jacobian of the chain from the moments to the elasticities, and aggregated by cluster or by survey design (`svyset`: PSUs, strata, finite-population correction); a design bootstrap of both stages; a diagnostic of the price signal before estimation; standard `e()` results and `estat`. Every variance is validated against the bootstrap under simple random sampling of clusters, sampling weights and a stratified design: the linearized standard errors are within 13 percent of the bootstrap for all 115 parameters. An option `compat` reproduces the Stata code published with Deaton (1997), checked against a line-by-line transcription of his programs; Appendix A records the three places where that code departs from the book and the effect of each.

**Keywords:** unit values, quality shading, price elasticities, household surveys, errors in variables, survey design, influence functions.

# 1 Introduction

Household surveys in developing countries rarely record prices, but they record expenditures and quantities, whose ratio, the unit value, carries the spatial variation of prices between villages. Deaton (1988) showed how to turn this into price elasticities of demand once two contaminations are removed: the unit value also reflects the quality chosen by the household, which rises with its income and falls when the price rises, and it is measured with an error correlated with the error in the quantity. His model, developed in Deaton (1990) and laid out in full in chapter 5 of Deaton (1997), remains the standard tool whenever price data are missing: a first stage within clusters, where prices are constant, identifies the income and household effects; a second stage between clusters, on the averages purged of those effects, identifies the price responses; weak separability of preferences turns the quality elasticity with respect to income into the quality response to price; and an errors-in-variables correction uses the first-stage residuals to purge the between-cluster moments.

The software has not followed the method. Deaton (1997) published the Stata programs of the Maharashtra and Pakistan applications — `allindia.do`, `mkmats.do`, `bootall.do` — as an appendix on the World Bank’s site; they are Stata 4 code tied to one data set, without weights, and their only standard errors come from a bootstrap of the second stage alone. The WELCOM package (Araar and Olivieri, 2018) wrapped that code into one command, `duvm`, and added sampling weights, a selection correction and results by group; its formulas are those of the published programs. We know of no other packaged implementation in Stata, R or Python; the applied literature re-implements the method from the paper each time, or falls back on unit values used as prices.

`duvm` 1.0 is a new implementation of the model as the book states it, in Mata, with the inference the method deserves. Section 2 states the model and the parameters that are identified. Section 3 gives the estimator, stage by stage, with the exact formulas used, including the treatment of weights. Section 4 derives the linearized variance and describes the bootstrap. Section 5 presents the diagnostic. Section 6 documents the command and reports the results on the four cereal groups of the Mexican ENIGH 2014, which ship with the package. Section 7 validates the variance. Appendix A documents the differences between the published code and the book, with their measured effect, and the option that reproduces the published code.

## 2 The model

Households  $h$  live in clusters  $c$ ; within a cluster all households face the same prices  $\pi_{Hc}$  of the  $M$  goods  $H = 1, \dots, M$ . Household  $h$  has total expenditure  $x_{hc}$ , characteristics  $z_{hc}$ , and for each good  $G$  a budget share  $w_{Ghc}$  and, when it buys the good, a unit value  $v_{Ghc} = \text{expenditure} / \text{physical quantity}$ . Following Deaton (1997, eqs. 5.76–5.77),

$$w_{Ghc} = \alpha_G^0 + \beta_G^0 \ln x_{hc} + \gamma_G^{0'} z_{hc} + \sum_{H=1}^M \theta_{GH} \ln \pi_{Hc} + f_{Gc} + u_{Ghc}^0, \quad (1)$$

$$\ln v_{Ghc} = \alpha_G^1 + \beta_G^1 \ln x_{hc} + \gamma_G^{1'} z_{hc} + \sum_{H=1}^M \psi_{GH} \ln \pi_{Hc} + u_{Ghc}^1. \quad (2)$$

The share equation has a cluster effect  $f_{Gc}$  uncorrelated with prices; the unit-value equation has none, since conditional on price a unit value reflects only quality and measurement error.  $\beta_G^1$  is the expenditure elasticity of quality;  $\psi_{GH}$  is the response of the unit value to prices, unity on the diagonal in the absence of quality shading;  $\theta_{GH}$  is the response of the budget share. The error terms include the measurement errors in quantities and unit values, and  $u^0$  and  $u^1$  are correlated within a household.

The quantities of interest follow from the identities  $w = qv/x$  and  $\ln v = \ln \pi + \ln \xi$ , where  $\xi$  is quality. Writing  $\bar{w}_G$  for the mean budget share,

$$e_G = 1 - \beta_G^1 + \beta_G^0 / \bar{w}_G \quad (3)$$

is the total expenditure elasticity of quantity, and the matrix of price elasticities of quantity is

$$E = -\Psi + D(\bar{w})^{-1}\Theta, \quad (4)$$

(Deaton, 1997, eqs. 5.88–5.89). Neither  $\Theta$  nor  $\Psi$  is identified from the data alone: the between-cluster regression of shares on unit values estimates only  $B = (\Psi')^{-1}\Theta'$ , eq. (5.86). Weak separability of preferences within each group supplies the missing restriction: the quality of good  $G$  depends on the price of  $H$  only through the quantity of  $G$ , so that

$$\psi_{GH} = \delta_{GH} + \frac{\beta_G^1 e_{GH}}{e_G}, \quad \Psi = I + D(\beta^1) D(e)^{-1} E, \quad (5)$$

eqs. (5.75) and (5.87). Combining (4) and (5) with  $\Theta = B'\Psi$  gives

$$\Psi = [I - D(\zeta)B' + D(\zeta)D(\bar{w})]^{-1}, \quad \Theta = B'\Psi, \quad E = [D(\bar{w})^{-1}B' - I]\Psi, \quad (6)$$

where the quality parameter of good  $G$  is

$$\zeta_G = \frac{\beta_G^1}{(1 - \beta_G^1)\bar{w}_G + \beta_G^0} = \frac{\beta_G^1}{\bar{w}_G e_G}, \quad (7)$$

eqs. (5.90)–(5.92). The product  $\zeta_G \bar{w}_G = \beta_G^1 / e_G$  is the share of a price change absorbed by quality shading; when it is small,  $\Psi$  is close to the identity and the unit values behave as prices.

**Completing the system.** The  $M$  goods with unit values do not exhaust the budget. Adding-up and homogeneity extend  $\Theta$  to an  $(M+1) \times (M+1)$  matrix  $\Theta^x$  whose last column is  $-\sum_H \theta_{GH} - \beta_G^0$  and whose last row makes every column sum to zero, eqs. (5.93)–(5.94);  $\bar{w}$ ,  $\beta^0$  extend by adding-up, and the composite good receives an assumed quality elasticity  $\beta_{M+1}^1$  (0.25 in Deaton's code; `qother()`). With  $\zeta^x$  from (7) on the extended vectors,

$$\Psi^x = [I + D(\zeta^x)D(\bar{w}^x)]^{-1}[I + D(\zeta^x)\Theta^x], \quad B^{x'} = \Theta^x(\Psi^x)^{-1}, \quad E^x = [D(\bar{w}^x)^{-1}B^{x'} - I]\Psi^x, \quad (8)$$

eqs. (5.95)–(5.97). The  $M \times M$  block of  $E^x$  equals  $E$ : completion adds the composite good without disturbing the others.

**Symmetry.** Slutsky symmetry requires  $\theta_{GH} + \beta_G^0 \bar{w}_H = \theta_{HG} + \beta_H^0 \bar{w}_G$ , eq. (5.71). Because it is nonlinear in  $B$  once quality enters, Deaton imposes it approximately, on  $B + \bar{w}\beta^{0'}$ , which coincides with the exact restriction when  $\Psi = I$ . With  $R$  the matrix that picks the  $M(M-1)/2$  pairs of  $\text{vec } B$  and  $r = -R(\beta^0 \otimes \bar{w})$ , the restricted estimate is

$$\text{vec } \tilde{B}_R = \text{vec } \tilde{B} + (I \otimes A)^{-1} R' [R(I \otimes A)^{-1} R']^{-1} (r - R \text{vec } \tilde{B}), \quad (9)$$

with  $A$  the corrected moment matrix of the second stage, eq. (5.102). `duvm` reports the elasticities with and without the restriction.

### 3 Estimation

#### 3.1 First stage: within clusters

Each of the  $2M$  equations is estimated by weighted least squares on the variables demeaned within cluster (the fixed-effect estimator), the unit-value equations on the households that report a unit value, the share equations on all households. With household weights  $\omega_h$ , the cluster mean of a variable is  $\sum_{h \in c} \omega_h y_h / \sum_{h \in c} \omega_h$  and the estimator is  $\hat{\beta} = (\tilde{X}' \Omega \tilde{X})^{-1} \tilde{X}' \Omega \tilde{y}$ . The first stage yields  $\hat{\beta}_G^0$ ,  $\hat{\beta}_G^1$ , the coefficients of the characteristics, and the residuals  $e_{Ghc}^0$ ,  $e_{Ghc}^1$ . From the residuals,

$$\hat{\omega}_G = \frac{\kappa_G}{n_G^+ - k - C_G^+} \sum_h \omega_h m_{Gh} (e_{Gh}^1)^2, \quad \hat{\chi}_G = \frac{\kappa_G}{n_G^+ - k - C_G^+} \sum_h \omega_h m_{Gh} e_{Gh}^0 e_{Gh}^1, \quad (10)$$

are the variance of  $u^1$  and the covariance of  $u^1$  and  $u^0$ , eqs. (5.81)–(5.82), where  $m_{Gh}$  marks the reporters,  $n_G^+$  and  $C_G^+$  count the reporting households and clusters,  $k$  the regressors, and  $\kappa_G = n_G^+ / \sum \omega_h m_{Gh}$  normalises the weights as Stata’s analytic weights do. The purged series

$$\hat{y}_{Ghc}^0 = w_{Ghc} - x_{hc}' \hat{\beta}_G^0, \quad \hat{y}_{Ghc}^1 = \ln v_{Ghc} - x_{hc}' \hat{\beta}_G^1, \quad (11)$$

keep the cluster effects and the price terms, eqs. (5.78)–(5.79);  $x_{hc}$  collects log expenditure, log household size and the characteristics.

#### 3.2 Second stage: between clusters

The cluster averages  $\bar{y}_{Gc}^0$  (over all households) and  $\bar{y}_{Gc}^1$  (over the reporters), weighted like the first stage, are residualised on the constant and the region and survey-round dummies when these are given (the *purge*, unweighted across clusters). Their between-cluster covariances,

$$S_{GH} = \text{Cov}_c(\bar{y}_{Gc}^1, \bar{y}_{Hc}^1), \quad R_{GH} = \text{Cov}_c(\bar{y}_{Gc}^1, \bar{y}_{Hc}^0), \quad (12)$$

are computed pairwise over the clusters where both series are observed. They contain the averaged first-stage errors,  $\omega_G/n_{Gc}^+$  and  $\chi_G/n_{Gc}$ , which do not vanish when clusters are small. Deaton’s errors-in-variables estimator removes them with the harmonic means of the cluster sizes, eq. (5.55):

$$\tilde{B} = (S - D(\hat{\omega})\bar{N}_+^{-1})^{-1} (R - D(\hat{\chi})\bar{N}^{-1}), \quad \bar{n}_G = \left( \frac{1}{C} \sum_c n_{Gc}^{-1} \right)^{-1}, \quad (13)$$

eq. (5.85). With weights, the variance of a weighted cluster mean of  $n$  independent errors is  $\sigma^2 \sum_h \omega_h^2 / (\sum_h \omega_h)^2$ , so the size of a cluster is its effective size  $n_c^{\text{eff}} = (\sum_h \omega_h)^2 / \sum_h \omega_h^2$  (Kish, 1965), which equals the number of households when the weight is constant within the cluster — the usual case, and the case of the Mexican data. This is the single point at which the weights change the second stage; the second stage itself treats each cluster as one observation, since each cluster is one draw of the price vector.

From  $\tilde{B}$ , the symmetry restriction (9) with  $A = S - D(\hat{\omega})\bar{N}_+^{-1}$ , then (7), (6) and (8) give the elasticities. Everything is closed-form; on 15,866 households, 2,622 clusters and four goods the whole chain takes a fifth of a second (Table 6).

#### 3.3 Non-buyers

A household that does not buy a good has a zero share and no unit value: its expenditure and its quantity are both zero. `duvm` calls a household a buyer when its budget share is positive, and reads a missing budget share as zero, so that the coding of non-purchase — a missing value or a zero — does not decide who is in the model: a household is never dropped because it does not buy one of the goods. The share equation is estimated on all households, the unit-value equation

on the buyers of the good; the purged unit-value average of a cluster without buyers is missing and the cluster drops out of the pairs that involve that good. No household or cluster is dropped for the other goods: each unit-value equation has its own buyers, and each between-cluster covariance its own clusters.

Survey files often hold a unit value for the non-buyers too, imputed from the buyers of the cluster or of a wider area so that every household has a price, or coded zero. For the model these are not observations. Option `nonbuyers(drop)`, the default, ignores them whatever their coding. Keeping imputed values (`nonbuyers(asis)`, or `nonbuyers(average)`, which imputes the mean of the buyers of the cluster) adds to the unit-value equation households whose unit value does not vary within the cluster: the quality elasticity  $\beta^1$  is biased toward zero, the more so the rarer the good, the residual variance  $\hat{\omega}$  is understated and the cluster sizes are inflated, so that the correction of (13) is too small. Values imputed from outside the cluster also give a price to clusters where nobody bought the good. In the example data, where the non-buyers carry such values, the quality elasticity of other cereals, bought by 16% of households, is 0.010 with them and 0.094 without them; the effect on the price elasticities is shown in Table 9. A code of zero is worse: a log unit value of zero is a price of one monetary unit, and the slope then measures the rate of purchase. `duvmdiag` counts each case (Section 5).

Option `csb(1)` adds to each share equation the inverse Mills ratio of a probit of buying the good on the first-stage regressors, the correction of the WELCOM version; the Mills ratio is treated as a fixed regressor in the variance. The unit-value equation, estimated on the buyers, describes the quality they choose; a selection correction of that equation is not implemented.

## 4 Variance

### 4.1 Linearization

The estimator is a smooth function  $\theta = g(\eta)$  of the vector of moments

$$\eta = (\text{vec } S, \text{vec } R, \hat{\omega}, \hat{\chi}, \hat{\beta}_{\cdot,1}^0, \hat{\beta}_{\cdot,1}^1, \bar{w}),$$

of dimension  $2M^2 + 5M$ , where  $\hat{\beta}_{\cdot,1}^j$  are the coefficients of log expenditure. The reported parameters  $\theta$  are every estimate displayed: the five price-elasticity matrices (unit values as prices; quality corrected, unrestricted and symmetry restricted,  $M \times M$  and completed), the expenditure elasticities,  $\beta^1$ ,  $\bar{w}$  and  $\zeta$  — 115 parameters for four goods. Then  $\hat{\theta} - \theta \approx G(\hat{\eta} - \eta)$  with  $G = \partial g / \partial \eta'$ , and  $\text{Var}(\hat{\theta}) = G \text{Var}(\hat{\eta}) G'$ .

**Influence functions.** Each component of  $\hat{\eta}$  is written as a sum of contributions,  $\hat{\eta} - \eta \approx \sum_c \Phi_c$ , one row per cluster. The first-stage moments are sums over households, attributed to their cluster: for the coefficients of an equation,  $\phi_h(\hat{\beta}) = (\tilde{X}'\Omega\tilde{X})^{-1}\tilde{x}_h\omega_h e_h$  on the households of that equation; for  $\hat{\omega}_G$ ,  $\kappa_G\omega_h m_{Gh}(e_{Gh}^1)^2/\text{df} - \hat{\omega}_G/n_G^+$ ; for  $\hat{\chi}_G$  the same with  $e^0 e^1$ ; for  $\bar{w}_G$ ,  $\omega_h(w_{Gh} - \bar{w}_G)/\sum \omega$ . The second-stage moments have their own cluster-level contribution,  $[(\tilde{y}_{Gc}^1 - m_G)(\tilde{y}_{Hc}^1 - m_H) - S_{GH}]/(n_{GH} - 1)$  over the  $n_{GH}$  clusters of the pair, and a generated-regressor term: the purged averages depend on the first-stage coefficients through  $\bar{y}_{Gc}^1 = \overline{\ln v_{Gc}} - \bar{x}_{Gc}'\hat{\beta}_G^1$ , so that

$$\frac{\partial S_{GH}}{\partial \beta_G^1} = -\frac{1}{n_{GH} - 1} \sum_c \tilde{x}_{Gc}(\tilde{y}_{Hc}^1 - m_H), \quad (14)$$

with  $\tilde{x}_{Gc}$  the cluster mean of the regressors over the reporters of  $G$ , purged like the  $y$ 's; the analogous terms hold for  $\partial S_{GH}/\partial \beta_H^1$  and for  $R_{GH}$  with respect to  $\beta_G^1$  and  $\beta_H^0$ . These derivatives multiply the cluster sums of the household influence functions of the coefficients. The purge and the harmonic cluster sizes are held fixed; the estimation of the purge coefficients is of second order for a residual covariance.

**Aggregation.** With clusters as the sampling units,  $\text{Var}(\hat{\eta}) = \frac{C}{C-1} \sum_c (\Phi_c - \bar{\Phi})'(\Phi_c - \bar{\Phi})$ . Under `vce(svy)` the rows are first summed within the PSUs of the design, which must contain the price clusters, and then

$$\text{Var}(\hat{\eta}) = \sum_s (1 - f_s) \frac{n_s}{n_s - 1} \sum_{p \in s} (\Phi_p - \bar{\Phi}_s)'(\Phi_p - \bar{\Phi}_s), \quad (15)$$

over strata  $s$  with  $n_s$  PSUs and finite-population correction  $f_s$ , as `svy` does for a total. The weights of the households, if the design has them, are the weights of the estimation.

**Jacobian.**  $G$  is computed analytically, by forward differentiation of the chain (13), (9), (7), (6), (8): a direction  $d\eta$  is propagated through  $dA = dS - D(d\omega)\tilde{N}_+^{-1}$ ,  $d\tilde{B} = A^{-1}(dR_f - dA\tilde{B})$ , the differential of the projection (9) including that of  $(I \otimes A)^{-1}$ ,  $d\zeta$ ,  $d\Psi = -\Psi d(\Psi^{-1})\Psi$ ,  $d\Theta$ ,  $dE$ , and the same for the completed system, for the unrestricted and the restricted  $B$ . Applying it to the  $2M^2 + 5M$  unit directions gives  $G$  exactly, at the cost of a few microseconds per direction; the test suite checks it against central differences to  $10^{-5}$  in relative terms.

## 4.2 Bootstrap

`vce(bootstrap)` draws clusters — or the PSUs of `svyset`, within strata — with replacement, lets the households travel with their cluster and their weights, treats a cluster drawn twice as two clusters, and re-runs both stages on every replication; the variance is that of the replications. Deaton’s programs bootstrap the second stage only, on the cluster-level file, holding the first stage fixed to save computer time (Deaton, 1997, p. 313); `shortcut` reproduces this. The shortcut assigns no variance to the expenditure and quality elasticities and omits the first-stage contribution to the price elasticities; Table 8 measures it.

## 5 Diagnostics

The method lives on the between-cluster variation of the unit values after the measurement-error correction. `duvmdiag`, with the syntax of `duvm`, runs the two stages without the variance. It first reports how the non-buyers are coded, good by good: the buyers, the budget shares coded missing (read as zero), the non-buyers that hold a unit value in the data — imputed, or coded zero — and the clusters that have a unit value but no buyer, whose price can only come from outside the cluster; each case is a warning. It then reports for each good the households and clusters whose unit value enters the model, the harmonic cluster sizes  $\bar{n}^+$  and  $\bar{n}$  that enter (13), the between-cluster standard deviation of the purged unit value, the share of its variance that the correction  $\hat{\omega}/\bar{n}^+$  removes, the corrected variance, the quality elasticity  $\beta^1$ , the expenditure elasticity and the quality share  $\beta^1/e$ ; then the condition number and smallest eigenvalue of the corrected moment matrix and the relative move that symmetry imposes on  $B$ . It warns when few households report, when  $\bar{n}^+ < 2$ , when the correction removes more than half of the variance or makes it non-positive, when  $\beta^1 < 0$  (the quality correction then raises the price elasticities instead of lowering them), when  $e$  is near zero ( $\zeta$  is then unstable), when the moment matrix is ill-conditioned, and when symmetry moves  $B$  by more than half. The same report is available after estimation with `estat diagnostics`.

## 6 The command

### 6.1 Installation

`duvm` requires Stata 14.2 or later and nothing else. The package is installed from its GitHub repository; the address below points to release 1.0.0, the version described in this note:

```
. net install duvm, from("https://raw.githubusercontent.com/aabdd12/duvm/v1.0.0") replace
```

It installs `duvm`, `duvmdiag`, the post-estimation commands, the help files (`help duvm`, `help duvmdiag`) and the dialog box. The example data of Section 6.4 and a guided tour of the command are copied into the current folder by the first line below; the second runs the tour, which loads the data:

```
. net get duvm, from("https://raw.githubusercontent.com/aabdd12/duvm/v1.0.0") replace
. do duvm_tour
```

Replacing `v1.0.0` by `main` in the address installs the latest version. The tables displayed with `stars` and written by `saveres()` are built by a private copy of `tabstars`, shipped with the package, so that no other module is needed. The replication files of the tables and figures of this note are in the folder `replication` of the repository; `master.do` runs them all.

## 6.2 Syntax

```
duvm goods [if] [in] [weight], hysize(varname) expend(varname) cluster(varname)
[region(varname) subround(varname) indcat(varlist) indcon(varlist)
csb(1) qother(#) nosymmetry nonbuyers(drop|average|asis)
vce(vcetype) level(#) hgroup(varname) dec(#) seall stars
saveres(filename) notable compat]
```

For each item  $g$  of *goods* the data hold  $wg$ , the budget share, and  $lvvg$ , the log unit value; a buyer is a household with a positive share, and `nonbuyers()` says what to do with the unit values of the others (Section 3.3). `vce()` is `cluster` (linearized, the default), `svy` (linearized, design from `svyset`), `bootstrap` with sub-options `reps()`, `seed()`, `strata()`, `psu()`, `svy`, `shortcut`, or `none`. `hgroup()` re-estimates the model within each group and reports the own-price elasticities by group. `stars` displays the tables with significance stars, each followed by its standard errors, and `saveres()` writes them to a Word, L<sup>A</sup>T<sub>E</sub>X, Excel, CSV or Markdown file (the extension gives the format). The WELCOM syntax — `hweight()`, `boot()` — is accepted. After estimation, `estat diagnostics`, `estat elasticities` (with `unrestricted`, `noquality`, `uncompleted`) and `estat quality` redisplay the results; `predict` gives the three Engel curves of the first stage — the budget share, the log unit value and the log quantity, with `stdp` for their linearized standard errors — and `estat engel` draws them; `e(b)` holds  $E^x$  row by row,  $e^x$  and  $\beta^1$ , `e(V)` their variance, and every intermediate matrix is stored. A dialog box is available with `db duvm`.

## 6.3 Engel curves after estimation

The first stage is linear in  $\ln x$  within clusters, eqs. (1) and (2). Holding the other regressors at their weighted means and the prices at their common level — which the cluster effects absorb — it traces three Engel curves for each good:

$$w_G(\ln x) = \bar{w}_G + \hat{\beta}_G^0 (\ln x - \bar{\ell}_G), \quad (16)$$

$$\ln v_G(\ln x) = \overline{\ln v}_G + \hat{\beta}_G^1 (\ln x - \bar{\ell}_G^+), \quad (17)$$

$$\ln q_G(\ln x) = \ln w_G(\ln x) + \ln x - \ln v_G(\ln x), \quad (18)$$

where  $\bar{w}_G$  and  $\bar{\ell}_G$  are the weighted means of the share and of  $\ln x$  over the estimation sample, and  $\overline{\ln v}_G$ ,  $\bar{\ell}_G^+$  the same means over the households that report a unit value. The first curve is the Engel curve of the budget share; the second shows how richer households buy dearer varieties of the good, with slope the quality elasticity  $\beta_G^1$ ; the third is the quantity bought, whose slope  $1 + \beta_G^0/w_G - \beta_G^1$  is the expenditure elasticity of quantity of eq. (3), now varying along the curve with the share. `predict` returns them with the options `share`, `quality` and `quantity`; option

**asobserved** gives instead the fitted value of each household, with its own characteristics and the effect of its cluster, which is what the first stage fits.

The quantity has a unit only if the unit value has one:  $q = wx/v$  is the expenditure of the household on the good divided by its unit value, the quantity bought by the household over the period of  $x$ , in kilograms when  $v$  is a price per kilogram. Two conditions follow. The variable given as total expenditure must be the denominator of the budget shares, or the level of the curve is shifted by the ratio of the two. And when the unit value of a group of items is an index, as in the example data,  $q$  is a quantity index whose level has no physical unit. The slope, an elasticity, is free of units in every case. Option **normalize** therefore reports  $\ln q_G(\ln x) - \ln q_G(\bar{\ell}_G)$ , the percentage difference in quantity from the household at mean expenditure, which compares goods on one scale.

The share curve is a straight line in  $\ln x$ , the Working-Leser form of Deaton's first stage. Far from the mean it can leave the unit interval: where it is not positive the log quantity is not defined and is left missing. The curves of a rank-three demand system such as EASI (Lewbel and Pendakur, 2009) can bend where these cannot; within the bulk of the distribution the two should agree.

**Standard errors.** Each curve depends on six statistics per good: the two means  $\bar{w}_G$  and  $\ln \bar{v}_G$ , the two means of  $\ln x$ , and the two within-cluster slopes. Their influence functions are those of Section 4 —  $\omega_h(y_h - \bar{y}) / \sum \omega$  for a mean,  $[(\tilde{X}'\Omega\tilde{X})^{-1}\tilde{x}_h\omega_h e_h]_1$  for a slope — and they are aggregated on the design of the estimates: by cluster, or by PSU within strata with the finite-population correction under **vce(svy)**. The covariances of the shares and slopes so obtained are those of **e(V)**, to  $10^{-15}$  in the tests. Option **stdp** applies the delta method to (16)–(18); after **vce(bootstrap)** the curves use the linearization on the bootstrap design, and after **csb(1)**, whose share equations also hold the inverse Mills ratio, only the unit-value curve has a standard error. **estat engel** draws the curves of all goods, one panel per good, against the percentiles of total expenditure — the layout of the **easi** command, so that the two can be set side by side — or against  $\ln x$  with option **lnx**, with a confidence band.

## 6.4 Results on the Mexican cereal groups

The example data are the four cereal groups of the ENIGH 2014 — corn, wheat products, rice, other cereals — for 15,866 households in 2,622 primary sampling units, prepared as in the WELCOM manual: unit values as Laspeyres indices of the items within a group. The file also gives a unit value to the non-buyers — the average of their cluster, or of a wider area when the cluster has no buyer — which **duvm** ignores by default (Section 3, Non-buyers). The model has log expenditure (current income), log household size, the age of the head, and dummies for sex and education, with the rural/urban region purged from the cluster averages; the sampling weight is used as an analytic weight.

Table 1: The data behind each good: mean budget share, households and clusters reporting a unit value, harmonic cluster sizes of the reporters ( $\bar{n}^+$ ) and of all households ( $\bar{n}$ )

| Good          | Share (%) | Reporters | %    | Clusters | $\bar{n}^+$ | $\bar{n}$ |
|---------------|-----------|-----------|------|----------|-------------|-----------|
| Corn          | 2.32      | 13,607    | 85.8 | 2,591    | 3.45        | 4.15      |
| Wheat         | 1.50      | 12,584    | 79.3 | 2,589    | 3.12        | 4.15      |
| Rice          | 0.15      | 4,481     | 28.2 | 1,808    | 1.58        | 4.15      |
| Other cereals | 0.10      | 2,600     | 16.4 | 1,499    | 1.33        | 4.15      |

Table 1 shows what the method has to work with. Corn and wheat are bought by 86% and 79% of households, in almost every cluster; rice by 28% and other cereals by 16%, in 1,808 and 1,499 of the 2,622 clusters. The buyers of a cluster number 3.4 and 3.1 in harmonic mean for corn

and wheat, but 1.6 and 1.3 for rice and other cereals: for these two goods the measurement-error correction rests on one or two households per cluster, and `duvmdiag` says so. Table 2 gives the first-stage parameters. Richer households pay more per unit for corn ( $\beta^1 = 0.019$ ), a little more for wheat (0.007) and much more for other cereals (0.094), a group of heterogeneous products; for rice the quality elasticity is  $-0.022$  with a standard error of 0.015, no quality upgrading — rice is a homogeneous product, and richer households may buy it in larger, cheaper packages. The expenditure elasticities of quantity run from 0.24 for corn to 1.07 for other cereals. The quality share  $\beta^1/e$  is 8% for corn and 9% for other cereals, and the diagonal of  $\Psi$  shows that the unit values respond to prices almost one for one.

Table 2: First stage: share and quality coefficients of log expenditure, expenditure elasticity of quantity, quality parameter  $\zeta$ , quality share  $\beta^1/e$  and own-price response of the unit value; linearized standard errors clustered by PSU

| Good          | $\hat{\beta}^0$ | $\hat{\beta}^1$ | (s.e.)   | $\hat{e}$ | (s.e.)  | $\hat{\zeta}$ | $\hat{\zeta}\bar{w} = \hat{\beta}^1/\hat{e}$ | $\hat{\psi}_{GG}$ |
|---------------|-----------------|-----------------|----------|-----------|---------|---------------|----------------------------------------------|-------------------|
| Corn          | -0.0171         | 0.0192          | (0.0047) | 0.242     | (0.020) | 3.418         | 0.079                                        | 0.936             |
| Wheat         | -0.0080         | 0.0069          | (0.0035) | 0.458     | (0.019) | 1.010         | 0.015                                        | 0.988             |
| Rice          | -0.0007         | -0.0223         | (0.0149) | 0.582     | (0.039) | -24.996       | -0.038                                       | 1.060             |
| Other cereals | 0.0002          | 0.0944          | (0.0251) | 1.066     | (0.055) | 88.435        | 0.089                                        | 0.933             |

Table 3: Price elasticities of quantity: quality corrected, symmetry restricted, completed system; rows are quantities, columns prices; linearized standard errors clustered by PSU in parentheses

| Quantity of     | $p_{\text{corn}}$ | $p_{\text{wheat}}$ | $p_{\text{rice}}$ | $p_{\text{other cereals}}$ | $p_{\text{all other}}$ |
|-----------------|-------------------|--------------------|-------------------|----------------------------|------------------------|
| Corn            | -0.814<br>(0.024) | 0.054<br>(0.013)   | -0.006<br>(0.012) | -0.007<br>(0.013)          | 0.531<br>(0.030)       |
| Wheat           | 0.079<br>(0.020)  | -0.820<br>(0.026)  | 0.102<br>(0.021)  | -0.003<br>(0.017)          | 0.183<br>(0.038)       |
| Rice            | -0.099<br>(0.187) | 1.001<br>(0.202)   | -1.572<br>(0.229) | -0.159<br>(0.058)          | 0.248<br>(0.266)       |
| Other cereals   | -0.178<br>(0.309) | -0.051<br>(0.250)  | -0.244<br>(0.088) | -0.760<br>(0.088)          | 0.167<br>(0.363)       |
| All other goods | -0.003<br>(0.001) | -0.004<br>(0.000)  | -0.000<br>(0.000) | 0.000<br>(0.000)           | -0.770<br>(0.001)      |

Table 3 is the final matrix. Own-price elasticities are  $-0.81$  for corn and  $-0.82$  for wheat, with standard errors of 0.02 to 0.03,  $-1.57$  for rice (0.23) and  $-0.76$  for other cereals (0.09); the composite of all other goods, whose share is 96% of the budget, has an own-price elasticity of  $-0.77$  that is essentially its expenditure elasticity. The cross-price elasticities of rice and other cereals, whose shares are 0.15% and 0.10% of the budget, carry standard errors of 0.06 to 0.36: small shares divide small coefficients. Table 4 follows the own-price elasticities through the treatments: the quality correction lowers the corn elasticity from  $-0.87$  to  $-0.81$  and that of other cereals from  $-0.85$  to  $-0.79$ , in proportion to their quality shares; for rice, whose quality elasticity is negative, it goes the other way, from  $-1.35$  to  $-1.43$ , as `duvmdiag` warns. Symmetry moves rice to  $-1.57$ , and completion changes nothing, as it should.

Table 5 re-estimates the model within each quintile of income, as `hgroup()` does. The own-price elasticities of corn and wheat are determined to within 0.07–0.36 in four quintiles out of five, those of rice and other cereals only loosely. The fourth quintile breaks down for every good, with standard errors in the thousands: within that group the corrected moment matrix of the second stage is singular. The rare goods cause it — a fifth of the households leaves one or two buyers of rice or other cereals in a cluster, often none — and since  $B = \tilde{S}^{-1}\tilde{R}$  is estimated as a system, a singular row spreads to all goods; the standard errors say so, and `duvmdiag` warns,

Table 4: Own-price elasticities of quantity under the four treatments

| Good          | No quality correction | Quality, unrestricted | Quality, symmetric | Completed system |
|---------------|-----------------------|-----------------------|--------------------|------------------|
| Corn          | -0.870                | -0.814                | -0.814             | -0.814           |
| Wheat         | -0.838                | -0.829                | -0.820             | -0.820           |
| Rice          | -1.355                | -1.428                | -1.572             | -1.572           |
| Other cereals | -0.845                | -0.787                | -0.760             | -0.760           |

Table 5: Own-price elasticities by quintile of per capita current income, the model re-estimated within each quintile; linearized standard errors

| Group      | Corn   | (s.e.)      | Wheat   | (s.e.)       | Rice     | (s.e.)        | Other    | (s.e.)        |
|------------|--------|-------------|---------|--------------|----------|---------------|----------|---------------|
| Quintile_1 | -0.922 | (0.089)     | -1.047  | (0.077)      | -2.976   | (3.353)       | -0.859   | (1.037)       |
| Quintile_2 | -0.883 | (0.150)     | -0.866  | (0.085)      | -1.985   | (1.319)       | -0.797   | (0.511)       |
| Quintile_3 | -0.716 | (0.317)     | -0.782  | (0.359)      | -0.762   | (0.757)       | -0.198   | (1.132)       |
| Quintile_4 | -3.117 | (16541.707) | -52.039 | (552927.653) | -276.532 | (2988308.023) | -131.973 | (1428627.426) |
| Quintile_5 | -0.804 | (0.084)     | -0.567  | (0.072)      | -1.071   | (0.846)       | -1.333   | (0.438)       |

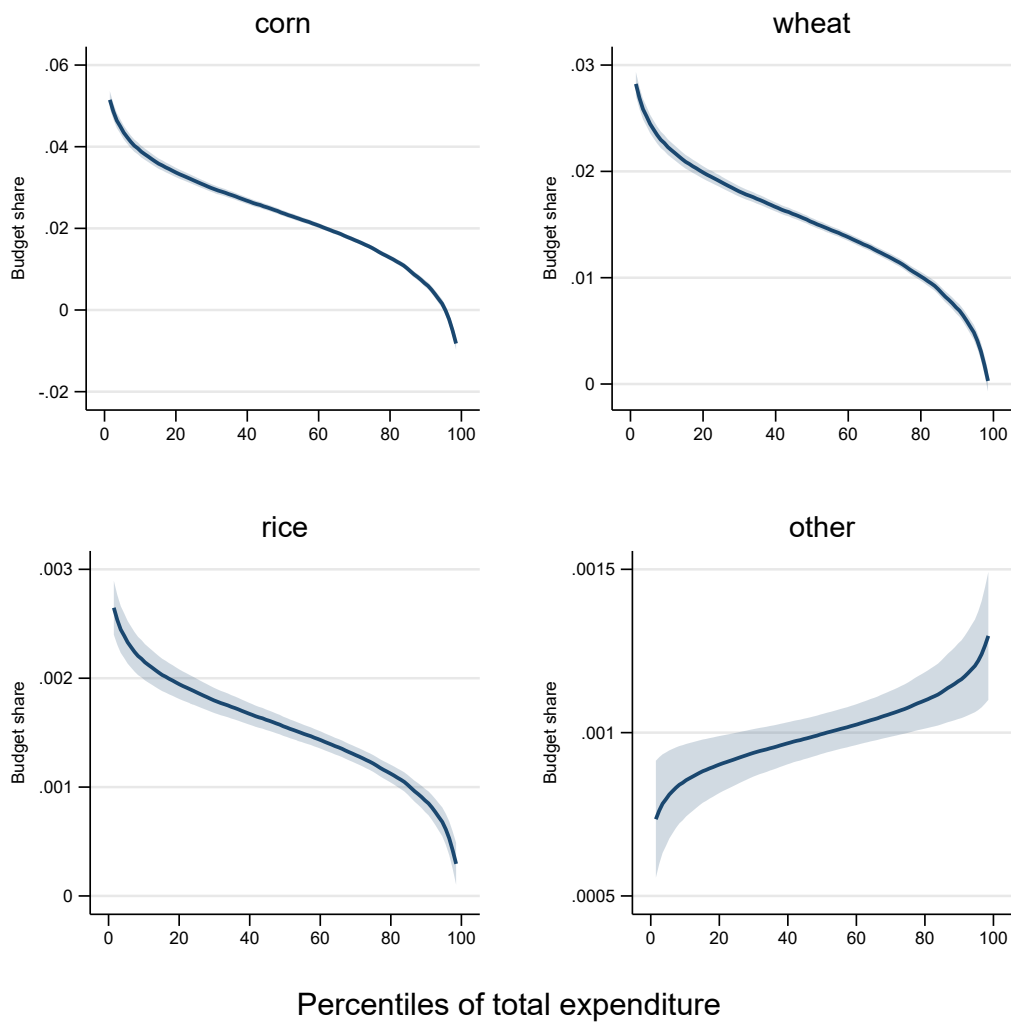
on every group, that the harmonic cluster size of the buyers falls below two. A group analysis of the unit-value model needs groups that keep the clusters whole — regions, not income classes — or a system without the rarest goods.

**Engel curves.** Figures 1–3 are drawn by `estat engel` after the model of Table 3, against the percentiles of total expenditure, with 95% linearized confidence bands. The curves are straight lines in  $\ln x$  (Section 6.3); against the percentiles they bend because  $\ln x$  is not spread evenly over the population.

The budget shares of corn, wheat and rice fall as expenditure rises: they are necessities, and the share of corn falls by two thirds between the 10th and the 80th percentile. The share of other cereals rises, a luxury with an expenditure elasticity above one. The bands are narrow for corn and wheat and wider for the two rare goods, where fewer households identify the slope. The limit of the linear form shows at the top: the share of corn reaches zero near the 96th percentile and that of wheat comes close to it at the 99th — 596 households, 3.8%, have a fitted corn share that is not positive. A share that flattens toward zero as expenditure grows is a curve a straight line cannot follow; there the Engel curves of a rank-three system such as EASI can bend, and within the bulk of the distribution the two approaches should agree.

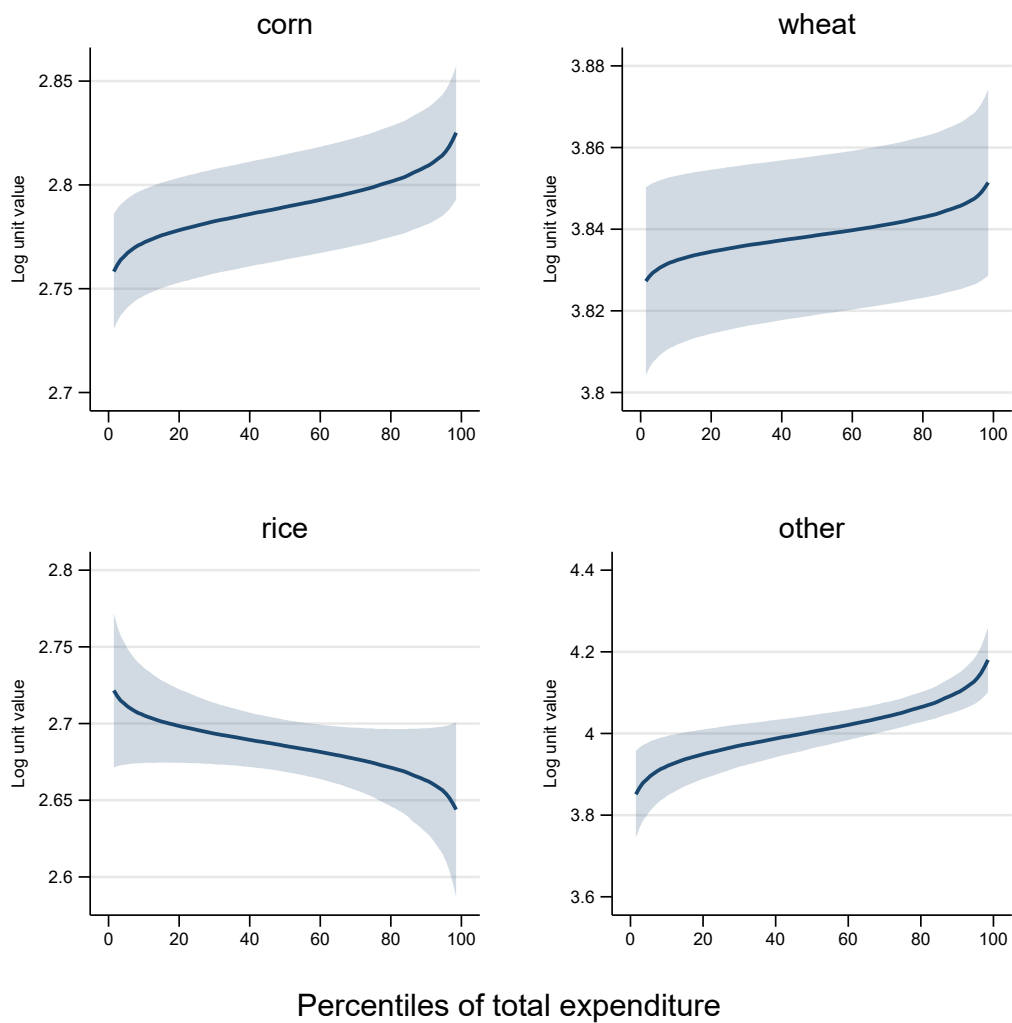
Figure 2 shows the quality chosen. Richer households pay more per unit for corn and, much more, for other cereals, whose unit value rises by 0.33 log points, about 40%, across the distribution, against 7% for corn and 2% for wheat. The unit value of rice falls slightly with expenditure, but its band contains a flat line: no quality upgrading. The bands are wide because they carry the uncertainty on the mean level of the unit value, which varies a lot between clusters; the slopes, which drive the quality correction, are known much more precisely than the width of the bands suggests (Table 2).

Figure 3 combines the two: the log quantity bought, set to zero at mean expenditure (option `normalize`), so that the four goods share one scale whatever the units of their unit values. Over most of the distribution the quantity rises with expenditure for every good, the slower the more necessary the good: from the 5th to the 90th percentile it is multiplied by 1.3 for corn, 2.6 for wheat, 3.5 for rice and 10.6 for other cereals. Corn peaks near the 70th percentile and falls beyond, and wheat falls in the top percentiles; this fall is not a finding but the linear share approaching zero, whose logarithm then drops without bound. The quantity curves are to be read in the bulk of the distribution, and the log quantity is left undefined where the fitted share is not positive.



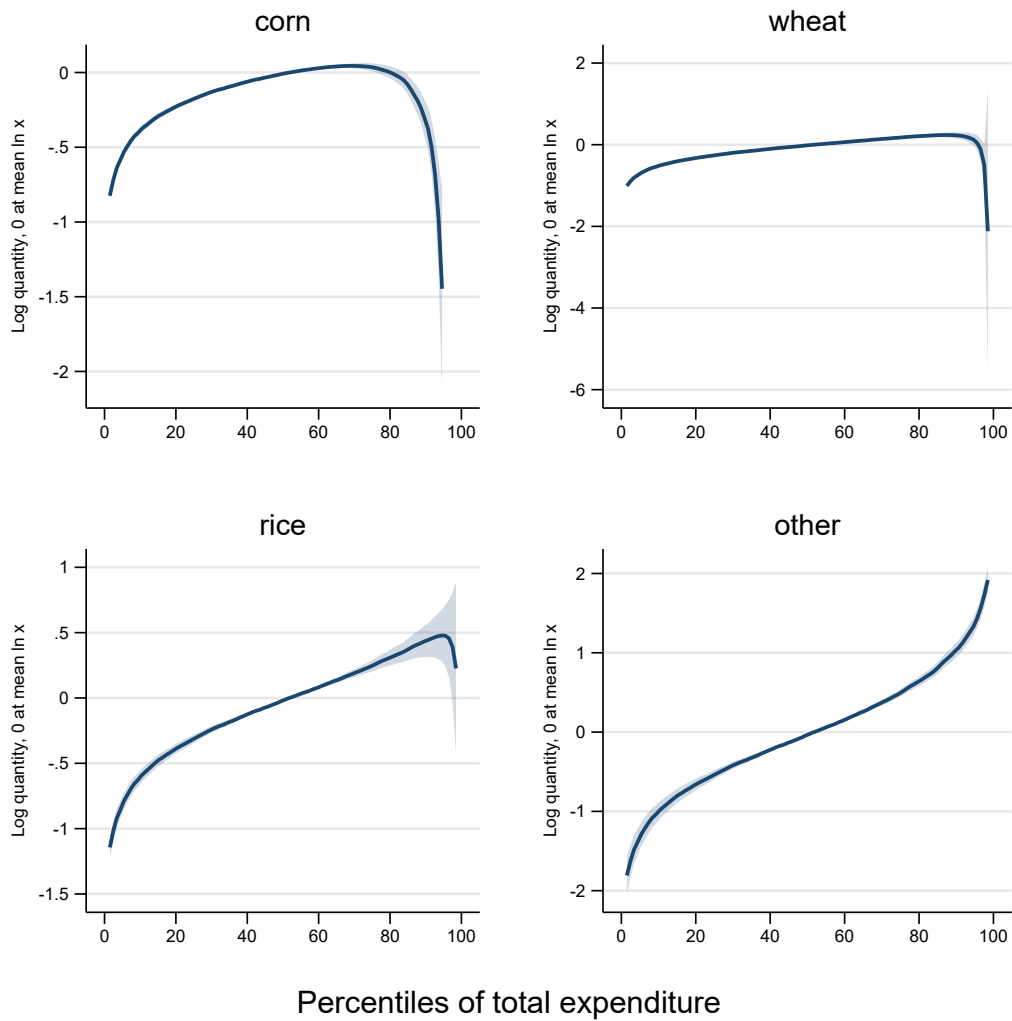
Budget share, at the means of the other regressors  
 straight lines in  $\ln x$  (the first stage of  $\text{duvm}$ ); tails trimmed at 1%; 95% confidence band (linearized)

Figure 1: Engel curves of the budget shares, at the means of the other regressors, against the percentiles of total expenditure; 95% linearized confidence bands



Log unit value, at the means of the other regressors  
straight lines in  $\ln x$  (the first stage of duvm); tails trimmed at 1%; 95% confidence band (linearized)

Figure 2: Engel curves of the log unit values (quality), at the means of the other regressors; 95% linearized confidence bands



Log quantity, 0 at mean  $\ln x$ , at the means of the other regressors  
 $\ln w + \ln x - \ln v$ ; falls where the share nears 0; tails trimmed at 1%; 95% confidence band (linearized)

Figure 3: Engel curves of the log quantities, normalized to zero at mean expenditure, at the means of the other regressors; 95% linearized confidence bands

## 6.5 Timing

Table 6: Execution times in seconds (StataNow 19.5, one core)

| Version                         | Estimation | Ten deciles | Bootstrap replication, both stages | Replication, shortcut |
|---------------------------------|------------|-------------|------------------------------------|-----------------------|
| WELCOM <code>duvm</code> (2017) | 2.35       | 5.60        | –                                  | 0.04                  |
| <code>duvm</code> 1.0           | 0.27       | 0.71        | 0.16                               | 0.03                  |

Table 6 compares the WELCOM version, whose loop over goods calls `areg` and `egen` and whose bootstrap re-reads files from disk, with the Mata engine. One estimation takes about a quarter of a second instead of 2.4 s; a bootstrap replication of both stages takes 0.16 s, so that 1,000 replications take under three minutes; Deaton’s shortcut, which repeats only the second stage, takes three hundredths of a second, against four for the WELCOM bootstrap that implements the same shortcut.

## 7 Validation of the variance

The linearized variance is validated against the bootstrap of both stages, 400 replications, in the order recommended for survey estimators: first with the clusters as the sampling units and no weights, then with the sampling weights, then with a stratified design declared to `svyset` (PSUs, strata = the quintiles of income, `pweight`). Table 7 reports the own-price standard errors under the two methods and, over the 115 parameters of  $\mathbf{e}(\mathbf{b})$  — every table, the shares, the expenditure, quality and  $\zeta$  parameters — the minimum, median and maximum of the ratio of the linearized to the bootstrap standard error. The median ratio is within one and a half percent of one in the three designs, and every ratio lies between 0.92 and 1.13, the spread expected from 400 replications.

Table 7: Linearized against bootstrap standard errors (400 replications of both stages): own-price elasticities, and the distribution of the ratio over the 115 parameters of  $\mathbf{e}(\mathbf{b})$

| Design                                                               | Standard error | Corn   | Wheat  | Rice   | Other  | All other | Ratio over all parameters |        |       |
|----------------------------------------------------------------------|----------------|--------|--------|--------|--------|-----------|---------------------------|--------|-------|
|                                                                      |                |        |        |        |        |           | min                       | median | max   |
| Simple random sampling of clusters                                   | Linearized     | 0.0214 | 0.0254 | 0.2003 | 0.0837 | 0.0011    | 0.940                     | 1.008  | 1.104 |
|                                                                      | Bootstrap      | 0.0207 | 0.0262 | 0.1815 | 0.0772 | 0.0011    |                           |        |       |
| Sampling weights ( <code>aweight</code> )                            | Linearized     | 0.0243 | 0.0259 | 0.2286 | 0.0880 | 0.0011    | 0.925                     | 0.988  | 1.122 |
|                                                                      | Bootstrap      | 0.0246 | 0.0266 | 0.2067 | 0.0870 | 0.0011    |                           |        |       |
| Survey design ( <code>svyset</code> , strata, <code>pweight</code> ) | Linearized     | 0.0243 | 0.0258 | 0.2285 | 0.0880 | 0.0011    | 0.944                     | 1.015  | 1.130 |
|                                                                      | Bootstrap      | 0.0233 | 0.0256 | 0.2113 | 0.0781 | 0.0011    |                           |        |       |

Table 8 measures what Deaton’s shortcut leaves out. Bootstrapping the second stage alone understates the standard error of the corn elasticity by 25% (ratio 1.34) and that of rice by 23% (ratio 1.30), that of the composite good by 10%, and those of wheat and other cereals by 5–7%. The first stage weighs most where its estimates are least precise relative to the price signal: for corn, whose quality share is the largest of the staples, and for rice, whose few buyers per cluster make the measurement-error correction large. The shortcut gives no standard error at all to the expenditure and quality elasticities.

## 8 Conclusion

`duvm` implements Deaton’s unit-value model as the book states it, with the inference the method had been lacking: a variance that accounts for both stages and for the survey design, a bootstrap of the whole procedure to check it, and a diagnostic of the price signal. The estimator is fast enough for resampling and for estimation by group. The formulas of the code published with the

Table 8: Bootstrap of both stages against the shortcut that resamples the cluster-level file only, 400 replications

| Bootstrap                    | Own-price elasticity |        |        |        |           | Expenditure elasticity |        |        |        |
|------------------------------|----------------------|--------|--------|--------|-----------|------------------------|--------|--------|--------|
|                              | Corn                 | Wheat  | Rice   | Other  | All other | Corn                   | Wheat  | Rice   | Other  |
| Both stages                  | 0.0249               | 0.0260 | 0.2149 | 0.0821 | 0.0011    | 0.0183                 | 0.0202 | 0.0380 | 0.0533 |
| Second stage only (shortcut) | 0.0186               | 0.0246 | 0.1656 | 0.0761 | 0.0010    | 0.0000                 | 0.0000 | 0.0000 | 0.0000 |
| Ratio                        | 1.337                | 1.058  | 1.298  | 1.079  | 1.114     | –                      | –      | –      | –      |

book remain available under `compat`, so that results computed with it can be reproduced and the effect of each correction can be seen; the appendix documents them.

## A The published code and the book

The Stata programs published with Deaton (1997) (`allindia.do` and `mkmats.do`, for the Maharashtra data; the book’s data were not distributed) implement chapter 5 with three departures from its equations.

1. **The quality parameter.** Program `mormat` computes  $\zeta_G = \beta_G^1 / (\beta_G^0 + 1 - \beta_G^1 \bar{w}_G)$ , with a 1 where eq. (5.92) has  $\bar{w}_G$ . Since  $\bar{w}_G$  is a few percent, this makes  $\zeta_G \approx \beta_G^1$  and reduces the quality correction to almost nothing; with the book’s formula  $\zeta_{\text{corn}} = 2.6$  against 0.015. The identity (5),  $\Psi = I + D(\beta^1)D(e)^{-1}E$ , holds to  $10^{-16}$  with the book’s  $\zeta$  and fails by 0.05 with the code’s.
2. **The completed system.** Program `complet` builds the extended  $\zeta^x$  as  $[\bar{w}(1 - \beta^1) + \beta^0] / \beta^1$ , the reciprocal of (7), through `syminv(diag(b1))`; `syminv` also zeroes any row whose pivot is not positive, so a good with a negative quality elasticity loses its correction in the completed system. This happens in subsamples: in the first decile of the Mexican data three of the four  $\beta^1$  are negative.
3. **The last row of  $\Theta^x$ .** `complet` appends the column sums of  $\Theta^x$  where eq. (5.94) requires their negatives, so that the columns of the completed matrix do not add to zero.

Two further details are matters of convention rather than error: the code obtains  $\hat{\chi}$  from an auxiliary within-cluster regression of  $e^1$  on  $e^0$  and the regressors rather than from (5.82), and the region of a cluster is that of its first household rather than the first one for which it is observed. The code is unweighted.

Table 9 moves from the published code to the book one correction at a time, on the unweighted model of Section 6. The quality parameter of eq. (5.92) alone changes little, because the completed system still inverts it. Once the completion uses  $\zeta^x$  itself, the quality correction becomes effective: the corn elasticity moves from  $-0.85$  to  $-0.83$  and the own-price elasticity of the composite good from  $-0.26$  to  $-0.74$ . The sign of the last row of  $\Theta^x$  touches only the composite good, which goes to  $-0.77$ ; the two conventions matter at the second or third decimal (rice moves from  $-1.14$  to  $-1.13$  with the region). These rows take the unit values as they are in the file, as the code would; the last row ignores those of the non-buyers, the default of `duvm` (Section 3). It matters more than any of the corrections for the rare goods: rice moves from  $-1.13$  to  $-1.55$  and other cereals from  $-0.72$  to  $-0.78$ , while corn and wheat do not move.

Option `compat` reproduces the published code, and `compatflags()` selects individual departures. The book’s data were not distributed, so the check is made on the Mexican data: the replication file `deaton_1997_ch5.do` transcribes `allindia.do` and `mkmats.do` line by line, keeping every formula and changing only the Stata 4 syntax, and Table 10 compares `duvm`, `compat` with it. The largest relative difference is below  $10^{-9}$ , with and without the region purge. With

Table 9: From the code published with Deaton (1997) to the book, one correction at a time: own-price elasticities of the completed, symmetry-restricted system (unweighted)

| Formulas                                             | Corn    | Wheat   | Rice    | Other   | All other |
|------------------------------------------------------|---------|---------|---------|---------|-----------|
| Deaton’s published code ( <code>compat</code> )      | -0.8464 | -0.8088 | -1.1514 | -0.7162 | -0.2551   |
| + $\zeta$ of eq. (5.92)                              | -0.8511 | -0.8101 | -1.1503 | -0.7175 | -0.2550   |
| + completion with $\zeta^x$ , not $1/\zeta^x$        | -0.8282 | -0.8169 | -1.1499 | -0.7130 | -0.7437   |
| + last row of $\Theta^x = -\text{colsum}$ (eq. 5.94) | -0.8282 | -0.8169 | -1.1499 | -0.7130 | -0.7692   |
| + $\chi$ from eq. (5.82)                             | -0.8287 | -0.8169 | -1.1445 | -0.7125 | -0.7692   |
| + region = first non-missing household (the book)    | -0.8236 | -0.8169 | -1.1254 | -0.7217 | -0.7688   |
| + unit values of non-buyers ignored (default)        | -0.8270 | -0.8192 | -1.5454 | -0.7752 | -0.7695   |

weights, which the code does not have, `compat` keeps its formulas and applies the weights as the default does.

Table 10: `duvm`, `compat` against a line-by-line transcription of the code published with Deaton (1997): largest relative difference (unweighted)

| Case          | Moments and coefficients | Elasticities |
|---------------|--------------------------|--------------|
| region purged | 8.0e-12                  | 2.8e-10      |
| no region     | 8.1e-12                  | 3.1e-10      |

## B Tests

The package ships with the scripts of the `replication/` folder, one per table of this note, run by `master.do`. The development test suite (not distributed) checks `compat` against the transcription of Deaton’s programs; checks the identities (5.87), (5.93), (5.94) and (5.98) on the default estimates to  $10^{-12}$ ; checks the analytic Jacobian against central differences; checks the linearized variance against the bootstrap under the three designs of Section 7; and exercises the bootstrap options, `hgroup()`, `estat` and `duvmdiag`.

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